Christian Garry

MSc Scientific Computing | Probability · Statistics · Optimisation | C++/Python
christiangarry.southafrica@gmail.com | +44 79 3232 6827 | christiangarry.com | linkedin.com/in/christian-tt-garry

Education

MSc Scientific Computing & Data Analysis (AI for Engineering) Durham, United Kingdom
Durham University Sep 2025 – Present

- Current average: Distinction (89%).
- **Focus:** probability, statistics, optimisation, numerical methods, Bayesian and regression models, HPC/GPU.

MEng Electronic Engineering Durham, United Kingdom
Durham University Sep 2020 – Jun 2024

- Upper Second-Class Honours (2:1).
- **Dissertation:** *Silicon Carbide JFET CPU* — simulation-driven CPU design with instruction-level verification and architectural trade-off analysis.

Experience

Industrial Tutor Durham, United Kingdom
Durham University, Department of Engineering Sep 2025 – Present

- Led weekly design tutorials, mentoring teams on feasibility-driven decision-making under cost, risk, and performance constraints.
- Assessed and provided structured feedback on technical feasibility, assumptions, and justification quality.

Graduate Communications Engineer Hebburn, United Kingdom
Siemens PLC Sep 2024 – Present

- Implemented and adapted C/C++ communication components (TCP/IP, serial) to support data exchange between simulated systems and PC-based engineering tools.
- Built Python-based engineering tooling to structure and query internal technical documentation and codebases using a retrieval-augmented workflow (local LLMs, vector search).
- Adapted existing network signal and waveform-generation utilities to support development and integration testing workflows.

Projects & Research

Silicon Carbide JFET CPU Durham, United Kingdom
Master's Dissertation Oct 2023 – Apr 2024

- Independently designed and simulated a 4-bit CPU using SiC JFET logic in LTspice, targeting computation in extreme temperature and radiation environments.
- Constructed a full verification toolchain (C++ compiler, assembler, Python automation) and validated instruction-level behaviour via emulator parity checks and waveform-based state reconstruction.
- Investigated architectural trade-offs, race conditions, and storage constraints, explicitly documenting assumptions, limitations, and scalability bottlenecks.

Key Skills

Probability & Statistics; Numerical Methods; Optimisation; Monte Carlo Simulation; Python; C++; High-Performance Computing; Linear Algebra

Leadership, Activities & Interests

Research Interests: Time-series modelling, signal validation, regime dependence in financial and physical systems. **Leadership:** Bishops Diocesan College Fencing Team Captain; Durham Fresher Representative (2022–23).
Activities: Durham University Fencing Team; Boxing (Student Fight Night).
Achievements: Weldon le Huray Fencing Scholarship (2020–24); South Africa U17 Fencing Champion; President's Award (Gold).